Math F113X: Homework Set 9

- Complete problems #5, 7, 9, 11, 13 - 17 from the Encryption section.
- Complete problems A, B, C, D, E, F.
- For extra credit, complete the Extra Credit Problem.

Problem A: Suppose you find the encrypted message below.

Znk qke oy zu ynolz he ykbkt. Yu znk rkzzkx g muky zu znk rkzzkx m.

(a) Fill out the frequency table below for the symbols in the encrypted text.

letter	frequency	letter	frequency	letter	frequency
A or a		J or j		S or s	
B or b		K or k		T or t	
C or c		L or l		U or u	
D or d		M or m		V or v	
E or e		N or n		W or w	
F or f		O or o		X or x	
G or g		P or p		Y or y	
H or h		Q or q		Z or z	
I or i		R or r			

(b) Identify the most frequent symbols and guess which letters they represent based on typical frequency tables of English text and supposing the encryption method is a substitution cipher. (See Example 5 from the text.)

frequency	1st	2nd	3rd	4th	5th
encrypted symbol					
likely/possible decrypted letter					

(c) Suppose the encryption method is a **Caesar (shift) cipher**. Determine the shift and decrypt the message.

Problem B: Decrypt the ciphertext ZOMRN PEBTB QIUA supposing that it was encrypted using a *progressive Caesar shift / shifting substitution cipher* with starting shift 6 (mapping $A \to G$) and shifts one additional space after each character is encoded.

Problem C:

(a) Encrypt the plaintext H U N C H B A C K O F N O T R E D A M E using a Vigenère cipher with keyword HUGO

(b) Decrypt the message R U J W W I L S H H S T R I F I U S supposing that it was encrypted with a Vigenère cipher with keyword CASE

Problem D: Double Transposition

(a) Encrypt the plaintext below using double transposition and first keyword WARM and second keyword MUSHY.
plaintext: WE PREDICT THE TRIPOD WILL FALL

(b) Decrypt the ciphertext below assuming double transposition was used where the first keyword was WARM and second keyword was MUSHY.
ciphertext: TAADP ENEBM TYNNR HEFES DEMEO UNCOW OR

Problem E: A message was encrypted using **double transposition** with the first keyword TRIM and second keyword TAR.

The first step of *decryption* was completed to give MWEOOCN. Answer the following about the **SECOND STAGE** of the decryption of this message.

1. What keyword should be used?
2. How many rows (other than the one for the keyword) will be in your table? How many entries in the last row of that table will be left blank?
3. Finish the decryption to recover the plaintext.

Problem F: Give brief answers to the following.

1. If a short message (say, 6-8 letters) must be sent securely, is a transposition cipher a good idea? Explain why or why not.
2. A long encrypted message is found to have E and T as its most frequent letters. Assuming the plaintext was in English, does this give a clue as to which encryption method might have been used? Explain.
3. Why is TABLES a better choice of keyword for a transposition cipher than ACCESS?
4. Why is BARK a better keyword for a *Vigenère cipher* than NOON?

Problem Extra Credit (5 points) **This is a challenging problem. Be patient with yourself.** Suppose you find the encrypted message below. Its letter frequency is in the following table.

RXVKWAMM METBB FTLA VX BTH WAMCARGYVK TV AMGTJBYMEFAVG XZ WABYKYXV, XW CWXEYJYGYVK GEA ZWAA AIAWRYMA GEAWAXZ; XW TJWYSKYVK GEA ZWAASXF XZ MCAARE, XW XZ GEA CWAMM; XW GEA WYKEG XZ GEA CAXCBA CATRATJBP GX TMMAFJBA, TVS GX CAGYGYXV GEA KXDAWVFAVG ZXW T WASWAMM XZ KWYADTVRAM.

letter	frequency	letter	frequency	letter	frequency
A	38	J	5	S	4
B	8	K	9	T	12
C	8	L	1	U	0
D	2	M	15	V	13
E	12	N	0	W	20
F	5	O	0	X	21
G	17	P	1	Y	14
H	1	Q	0	Z	9
I	1	R	6		

(a) Identify the most frequent symbols and guess which letters they represent based on typical frequency tables of English text and supposing the encryption method is a substitution cipher. (See Example 5 from the text.)

frequency	1st	2nd	3rd	4th	5th
encrypted symbol					
likely/possible decrypted letter					

(b) Based on your work in parts a and b, explain why this is **unlikely** to be a **Caesar cipher**.

(c) Fill in the decryption key below and decrypt the message. Note that to do this will require ad hoc and guess-and-check strategies.

encrypted symbol	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
decrypted symbol																										

(d) What aspects of the encryption method made it much easier to decrypt?

Remember to write up your homework solutions according to the homework writeup guidelines.

Homework is graded using the following rubric for each problem (or problem part):

2 points: You provided a complete answer, with supporting work, written up clearly

1 point: Some attempt at a solution, but incomplete writeup / unclear / illegible / no answer

1 point: Only an answer, with no supporting work

0 points: Missing.

After you do the homework, you need to check your answers against the solutions! Then figure out your errors (if any) and revise your homework before you submit it.